

Meftahul Jannat

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Research Interests

Machine Learning, Computer Vision, LLM, Bioinformatics, and Cyber Security

Education

B.Sc. in Electronics & Telecommunication Engineering Rajshahi university of Engineering & Technology (RUET) Result: CGPA 3.59/4.00 Position: 7th Thesis: Lung Segmentation with Lightweight Convolutional Attention Residual U-Net <i>Algorithm: CNN, Residual U-Net</i>	2020 – 2025
Higher Secondary Certificate Rajshahi College, Rajshahi Result: GPA 5.00/5.00	2018 – 2019
Secondary School Certificate Govt. Laboratory High School, Rajshahi Result: GPA 5.00/5.00	2012 – 2017

Experience

Lecturer , Dept. of ICT, Bangladesh Army University of Science and Technology (BAUST), Saidpur	Jan 2025 – Current
- Delivered lectures and interactive coursework across core Information & Communication Technology domains, including Analog Communication & Radio-TV Engineering, Optical Fiber Communication, and Wireless & Mobile Communication, Data Communication & Computer Networks and Management Information Systems (MIS) - Supervised undergraduate final-year Capstone Projects, guiding students through problem formulation, theoretical modeling, simulation, hardware prototyping, and final thesis defense preparation.	
Trainee Engineer , IC Physical Design (PnR) – Ulkasemi Pvt. Limited	Dec 2025 – Jan 2026
- Execute Place and Route (PnR) flows under supervision, translating synthesized gate-level netlists into physical chip layouts using industry-standard EDA tools such as Cadence Innovus, Synopsys IC Compiler. - Develop automation scripts utilizing Tcl, Python, and Bash/Perl to streamline physical design tasks, report parsing, and EDA tool constraint management.	
Research Assistant , AIMS Lab – United International University	Nov 2025 – Nov 2025
- Conduct research on automated Scribe technology AIMScribe at AIMS Lab (UIU), architecting novel pipelines that integrate Speech-to-Text (STT), Pyannote speaker diarization, Vision-Language Models (VLMs), and LLMs for multimodal document generation. - Evaluated and experimented deep learning models across speech, vision, and NLP modalities, utilizing rigorous cross-validation and statistical evaluation metrics (WER, DER, BLEU/ROUGE scores) to validate model performance.	

Publications

Real-time jute leaf disease classification using an explainable lightweight CNN via a supervised and semi-supervised self-training approach	Oct 2025
<i>Meftahul Jannat</i> , Md Shahab Uddin, Mohammad Asif Hasan, Md Saimun Alam, Avijit Paul, Muhammad EH Chowdhury, Julfikar Haider <i>Frontiers in Plant Science</i> – DOI: 10.3389/fpls.2025.1647177	

Lung Segmentation with Lightweight Convolutional Attention Residual U-Net.

March 2025

Meftahul Jannat, Shaikh Afnan Birahim, Mohammad Asif Hasan, Tonmoy Roy, Lubna Sultana, Hasan Sarker, Samia Fairuz, Hanaa A Abdallah

Diagnostics – DOI: 10.3390/diagnostics15070854

Cotton Leaf Disease (CLD) Classification Using Multihead External Attention Vision Transformer.

Dec 2024

Meftahul Jannat, Hasan Sarker

27th International Conference on Computer and Information Technology (ICCIT)

DOI: 10.1109/ICCIT64611.2024.11022519

Projects

CXR-Agent: Vision-Language Model for Chest X-Ray Analysis.

github.com/meftahuljannat19/cxr-agent

- Finetune Vision LLM to generate medical reports from Chest X-ray images using Unsloth
- Finetuned different LLMs (Llama-3.2-11B-Vision-Instruct, Qwen2-VL-7B-Instruct) on a publicly available chest x-ray dataset (Chexpert, MIMIC)

Automated Toll Collection System using number plate recognition with YOLOv8 and EasyOCR on custom dataset annotated with CVAT

github.com/meftahuljannat19/Automated_Toll_System_YOLOv8

- Annotated a custom vehicle dataset of total 350 vehicle (car, bus, truck) using CVAT then trained YOLOv8 model on this custom dataset to detect different vehicles.
- The number plate is detected using edge detection and recognized with EasyOCR, enabling an automated toll payment system. The toll gate is then automatically controlled through a microcontroller, ensuring a seamless and efficient process.

Technologies

Languages: C++, C, Python, MATLAB, VHDL

Framework: PyTorch, Tensorflow, Keras, Unsloth

AI/ML Techniques: CNN, LSTM, NLP, Transformers, YOLO, LLM, Explainable AI, GAN

Development Tools: FastAPI

Hardware: Arduino, Raspberry Pi, ESP32

Software: MATLAB, COMSOL Multiphysics

Awards & Achievements

Research Publication Award

June 2025

RUET INNOVISTA 2025

References

Dr. Md. Kamal Hossain

Professor

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